

? Suppose you have been given a fair coin and you want to find out the odds of getting heads. Which of the following option is true for such a case



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- a. None of these ☐
- b. odds will be 0.5 ☐
- c. odds will be 1 ☒
- d. odds will be 0 ☐

A regression between foot length (response variable in cm, y) and height (independent variable in inches, x) for 33 students resulted in the following regression equation

$$y = 10.9 + 0.23x$$

? One student in the sample was 73 inches tall with a foot length of 29 cm. What is the residual (error) for this student

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- a. 0.00 cm ☐
- b. 1.31 cm ☒
- c. -1.31 cm ☐
- d. 29 cm ☐

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?Which statement is not true about gradient descent

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- a. Gradient descent is a first-order iterative optimization algorithm for finding a local minimum of a differentiable function ☐
- b. It is related to derivative of a function ☒
- c. It is used in many AI algorithms to optimize the cost function ☐
- d. During computation of the gradient descent the steps are taken proportional to the positive of the gradient (or approximate gradient) of the function at the current point ☐

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What is the name of this principle in machine learning: "The goal of machine learning is that the learned model should perform well on new samples, not just those on which the model has been trained"

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- a. Classification ☐
- b. Underfitting ☐
- c. Overfitting ☐
- d. Generalization ☒



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:An instructor gives the same data as given below to four students

(40,1200) ,(600 ,30) ,(400 ,20) ,(100 ,10) ,(1 ,1)

They each come up with four different answers for the straight line regression model. Only one is correct. The correct model is

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a. $y = 30x - 200$ ☐

b. $y = -139.43 + 29.684x$ ☒

c. $y = 60x - 1200$ ☐

d. $y = 1 + 22.782x$ ☐

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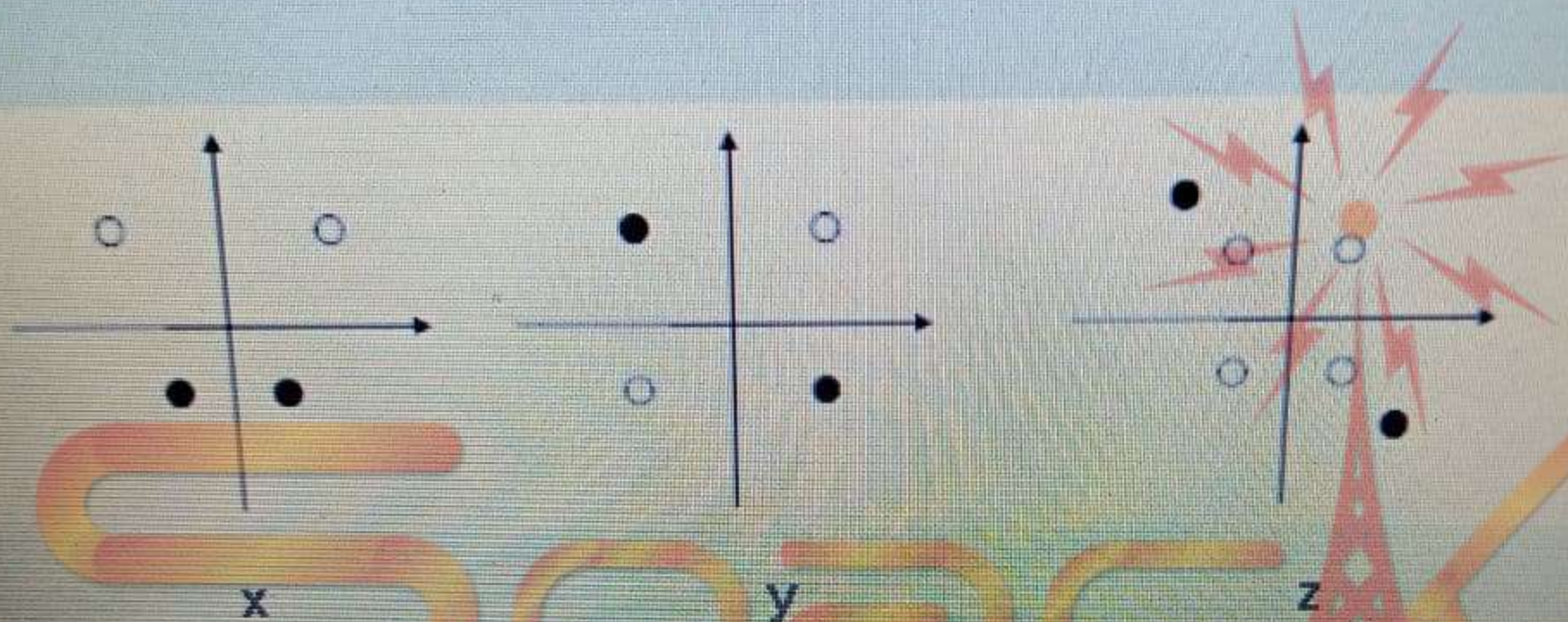
?The logit function(given as $l(x)$) is the log of odds function. What could be the range of logit function in the domain $x=[0,1]$



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- a. $(0,1)$ ☐
- b. $(-\infty, 0)$ ☐
- c. $(-\infty, \infty)$ ☒
- d. $(0, \infty)$ ☐

:A single layer neural networks (Perceptron) can learn



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a. x and y

b. z

c. y and z

d. x

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What is backpropagation(Neural network) ?

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- a. It is the transfer of outputs back from the hidden layer to the input layer using a set of recurrent connections ☒
- b. It is the transfer of error back through the network to allow the weights to be adjusted ☐
- c. It is the transfer of error back through the network using a set of recurrent connections ☐
- d. It is the transfer of error back through the network to adjust the inputs ☐

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:An example of a hyperparameter of a neural network

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a. Number of inputs ☒

b. Bias ☐

c. Learning rate ☐

d. Weights ☐

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In machine learning the simple linear regression equation can be written as $y = b + wx$

the parameter b represents the

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- a. estimated weight ☐
- b. independent variable ☐
- c. average or predicted response ☐
- d. estimated bias ☒

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During training neural network, epochs represent the total number of

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- .a. passes of the test data through the network ☐
- .b. input layer nodes ☐
- .c. network nodes ☐
- .d. passes of the training data through the network ☒

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?Why are linearly separable problems interesting to neural network researchers

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- a. Because they are the only mathematical functions that are continuous ☐
- b. Because they are the only mathematical functions that you can draw ☐
- c. Because they are the only problems that a perceptron can solve successfully ☒
- d. Because they are the only problems that a neural network can solve successfully ☐

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SOM is a supervised learning algorithm neural network

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صح ☐

خطأ ☒

In supervised learning, training set of data includes

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a. Input ☐

b. Both input and output ☒

c. Output ☐

d. None ☐

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: In neural network, one disadvantage of sigmoid activation function

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a. It is continuous ☐

b. easy to derive ☐

has a gradient vanishing phenomenon (its derivative is close to 0 at the point away from the central point) ☒

d. It is monotonic ☐